

Adapting the MEE Pipeline to PUNCH Coronagraph Data

Centroid detection, Gaia validation, and what the noise floor taught us

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Executive Summary

I adapted our MEE2024 Tab 1 centroid pipeline to PUNCH Level-3 white-light coronagraph images. PUNCH is different enough from eclipse data that it took three full pipeline iterations to get a clean result — the corona generates thousands of false detections, so I masked the inner field and validated everything that remained against the Gaia star catalog using PUNCH's mission pointing solution.

The pipeline works: of 4,343 detected centroids in the final run, 86% correspond to real catalog stars. The same analysis also quantified a hard limitation — at PUNCH's plate scale ($\sim 81''/\text{pixel}$), the gravitational deflection signal for even our closest detectable star is roughly 1,000x below our measured position noise, and moving closer to the Sun cannot fix it.

The real value is the machinery: centroid finding, catalog matching, and the deflection sensitivity check are now built and validated, and the numbers tell us exactly what instrument they need next — a finer plate scale that reaches closer to the Sun. For future runs I also plan a visual-first step (confirming stars by eye before trusting automated matches), a check I skipped on this first pass.

1. Data & Setup

INSTRUMENT

PUNCH (Polarimeter to Unify the Corona and Heliosphere), Level-3 white light.

FRAME

2026-02-19 00:16 UTC, 4096 x 4096 pixels, Sun centered at pixel (2048, 2048).

PLATE SCALE

~ 81 arcsec/pixel — about 44x coarser than our eclipse data ($\sim 1.85''/\text{px}$); $1 R_{\odot} \approx 12$ px.

THE CHALLENGE

the corona fills the inner $\sim 20\%$ of the frame with bright, lumpy structure that the centroid finder happily mistakes for stars.

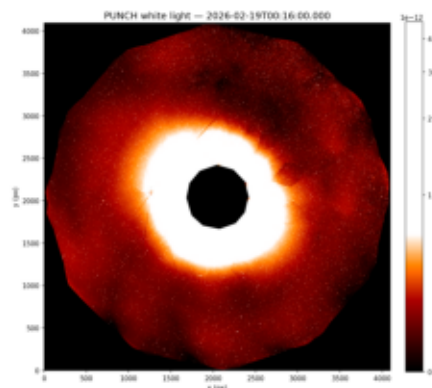


Figure 1. The 2026-02-19 PUNCH white-light frame. The Sun (black disk) is surrounded by the bright corona; real stars are faint points in the outer field. Everything inside the corona is a minefield for automated star detection.

RUN	THRESHOLD	CORONA HANDLING	CENTROIDS	OUTCOME
01	$\sigma = 6$	none (MEE's blob mask never fires on PUNCH)	9,255	mostly corona junk
02	$\sigma = 10$	none	4,824	still corona-dominated
03	$\sigma = 10$	circle mask, $r < 1000$ px	4,343	clean — used for analysis

MEE's built-in saturation mask assumes an over-exposed Sun; PUNCH's Sun is a dark hole, so that mask never activated. For run 03 I wrote my own pre-processing step: every pixel within 1,000 px of Sun-center is replaced with a single flat sky value (the median of a thin ring just outside the mask), blanking 18.7% of the image. This removes the corona's false bumps before the centroid finder ever sees them.

One important consequence: at 12 px per solar radius, a 1,000 px mask means the nearest star we can possibly detect sits at ~ 83 solar radii. That inner limit in everything below is self-imposed by the mask — not a physical property of the data.

2.1 Validating against Gaia

A detected centroid counts as a match if it lands within 2 pixels of where the Gaia catalog says a star should be (projected through PUNCH's pointing solution). Two numbers summarize the result:

- Purity = 86.3% — of the 4,343 things I flagged, 3,749 are real stars. The detector rarely cries wolf.
- Completeness = 44.8% — of the 8,560 catalog stars bright enough to be in the usable field, I recovered about 45%. The fainter half stay below PUNCH's noise.

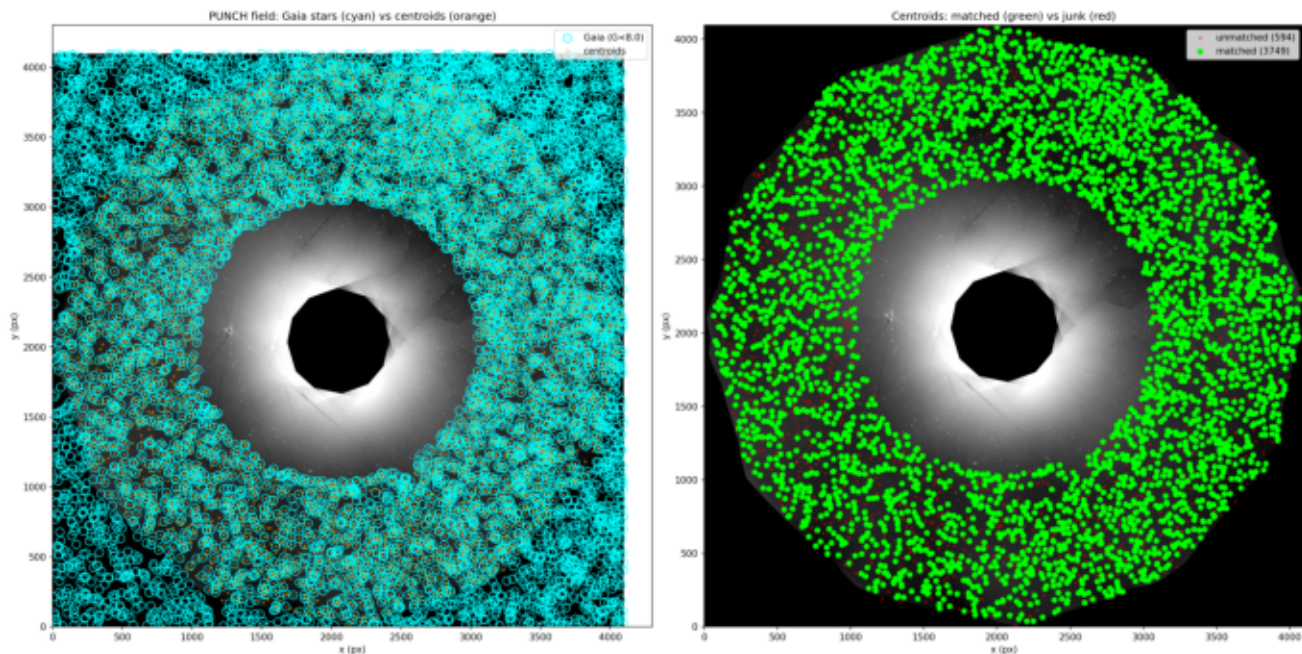


Figure 2. Run 03 validation. Left: detected centroids (cyan) on the masked PUNCH frame. Right: the Gaia truth set — green points are catalog stars recovered by the pipeline. The empty central region is the $r < 1000$ px corona mask.

83.7 $R_{\odot}$

closest detectable star
(set by the corona mask)

0.021"

predicted deflection there
 ≈ 0.0003 pixels

22.6"

measured position noise
 ≈ 0.28 pixels

$\sim 1,000\times$

signal below the noise
at every visible radius

The natural question is whether detecting stars closer to the Sun would close the gap. It would not. Deflection grows as $1.75''/r$, so to lift the signal above even a tenth of a pixel at PUNCH's scale, a star would need to sit at $r < 0.2$ solar radii — inside the Sun. The limitation is not where our stars are; it is that 81" pixels are too coarse for this measurement at any radius.

That is a useful thing to know precisely, and it points directly at what would work: the same pipeline pointed at a coronagraph with a finer plate scale, which can both resolve smaller shifts and see stars much nearer the limb.

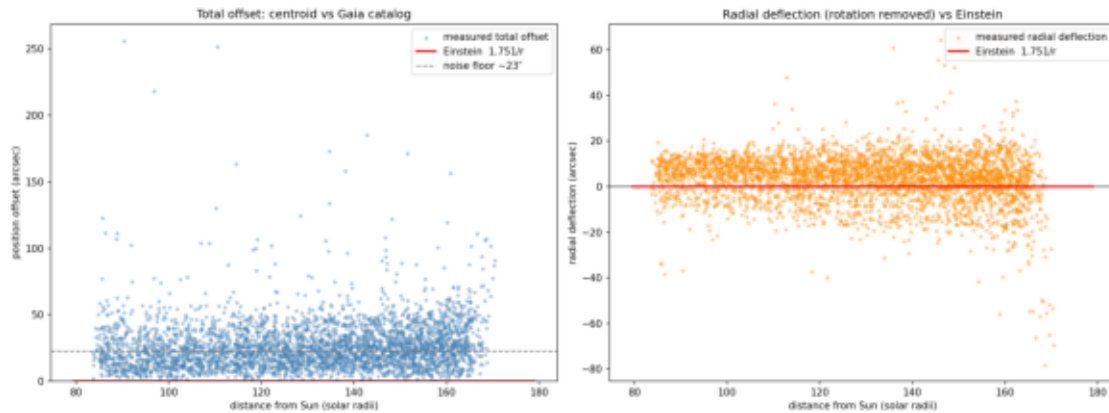


Figure 3. Measured star offsets vs. distance from the Sun (left: total offset, right: radial component). The red curve is the predicted Einstein deflection ($1.75''/r$) — three orders of magnitude below the measurement noise at every distance PUNCH can see. The dashed gray line is the $\sim 23''$ noise floor.

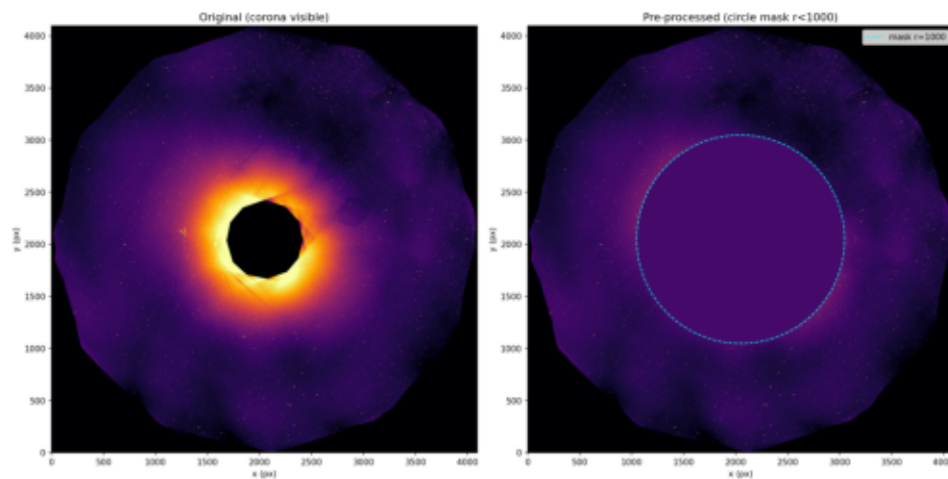


Figure 4. The corona mask that made run 03 work. Left: original frame. Right: the pre-processed input actually fed to the centroid finder — everything inside $r < 1000$ px (dashed circle) is flattened to a single sky value.

4 Refined Approach & Next Steps

4.1 A visual-first check

One lesson from the first pass: I trusted the automated brightest-detections list without first confirming by eye which of them were actually stars. A conversation with Henry Throop (NASA) reinforced a better habit — inspect the frame visually and confirm star-like sources before matching. (For this I use DS9, a standard free astronomy viewer for FITS images that lets you zoom and adjust the brightness stretch.)

As a first application, I built a "top-20" comparison: the 20 brightest detected centroids plotted against the 20 brightest predicted Gaia stars, with no automatic matching — pure visual inspection. It was immediately informative: several of the brightest detections near the Sun have no catalog counterpart at all (they are corona structure, not stars), while the nearest visually confirmed real star sits at ~ 98 solar radii, with other confirmed matches out to ~ 160 .

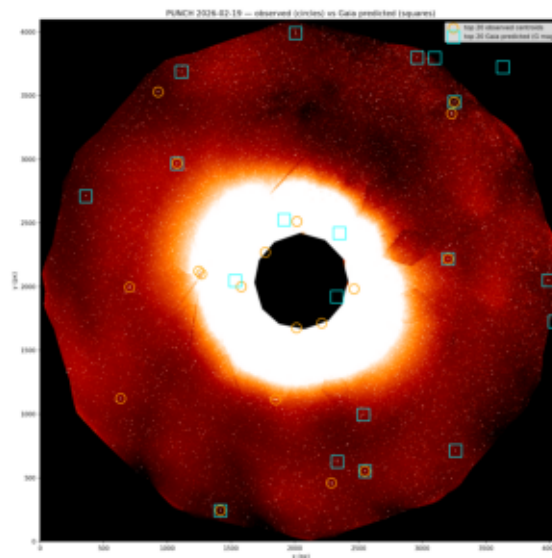


Figure 5. Top-20 brightest detections (orange circles) vs. top-20 predicted Gaia stars (cyan squares). Where a circle has no square, the "star" is corona. The closest genuine match, at roughly the 3 o'clock position, lies at ~ 98 solar radii.

4.2 A working principle I've adopted

For anything astrometric, I now treat the native mission FITS as the only source of truth. Display products (JP2s, PNGs) are stretched, 8-bit, and lossy-compressed — I verified that a JP2 re-wrapped as FITS carries only 256 distinct pixel values versus thousands in the real data — so they cannot be used for sub-pixel position work, no matter the file extension.

Next Steps

- Apply the visual-first workflow to establish a trusted star list per frame.
- Repeat the brightest-N analysis across many epochs, tracking the same stars frame to frame to pin down the real error budget. This can run on PUNCH as a method rehearsal, but the payoff comes on a finer-scale instrument — we've confirmed PUNCH itself cannot reach the signal at any radius.
- Source a science-grade frame from a finer-scale coronagraph (STEREO, at $\sim 15''/\text{px}$ and reaching within a few solar radii of the limb, is the leading candidate) — that is where this pipeline points next.